Species-Aware Module Review: Endogenous Protein Lipoylation in *Pseudomonas putida* KT2440

Taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Module:** Endogenous protein lipoylation (LipB octanoyltransferase → LipA lipoate synthase) **Local bucket:** kegg:ppu00785 "Lipoic acid metabolism" (cofactors_vitamins_redox), 19 primary genes **Date:** 2026-07-18 · Review complete (5 iterations)

---

1. Executive summary

The endogenous protein-lipoylation module is a **narrow, two-reaction biosynthetic pathway** that installs the lipoyl cofactor de novo on lysine residues of lipoyl-carrier domains: (1) a LipB-family octanoyltransferase transfers octanoyl from octanoyl-ACP onto the apo lipoyl domain, then (2) a LipA-family radical-SAM lipoate synthase inserts two sulfur atoms to yield the mature lipoyl group.

In *P. putida* KT2440 **both steps are covered and canonical**:

- **Step 1 = LipB (PP_4801 / Q88DM4)** — Octanoyltransferase, EC 2.3.1.181, carries the diagnostic PROSITE **LIPB** signature (PS01313) and the Octanoyltransferase InterPro family (IPR000544). A *bona fide* LipB, **not** an LplA-like LipM.
- **Step 2 = LipA (PP_4800 / Q88DM5)** — Lipoyl synthase, EC 2.8.1.8, radical-SAM enzyme (Pfam PF04055, IPR003698 Lipoyl_synth) with 4Fe-4S / SAM annotation.

The two genes are **genomically adjacent** (PP_4800–PP_4801), consistent with a lipA-lipB unit. A whole-proteome scan of taxon 160488 found **no LplA-family salvage ligase**, so de novo synthesis appears to be KT2440's sole route to protein-bound lipoate. However, a domain-based scan (correcting an earlier name-only search) revealed a **candidate LipM/LipL-family GcvH-relay octanoyltransferase, PP_0423** (InterPro IPR050664; PANTHER "octanoyl-[GcvH]:protein N-octanoyltransferase"). This is an **out-of-module** auxiliary per the module's scope (GcvH-relay architectures are excluded) and does not affect core satisfiability, but it is flagged for review.

Module recommendation: COVERED. LipB and LipA cover both steps. The remaining 17 of the 19 bucket genes are **out-of-module** (lipoyl-domain acceptors, non-lipoylated E1 subunits, E3 dehydrogenases, and glycine-cleavage T-proteins) and reflect KEGG map breadth rather than pathway steps; PP_0423 (not in the bucket) is a separate out-of-module relay candidate.

---

2. Target-organism pathway definition

In scope (the module): the enzymatic *installation* of the lipoyl cofactor on apo lipoyl domains, comprising exactly two reactions:

1. Octanoyl-ACP + apo-[lipoyl domain] → octanoyl-[lipoyl domain] + ACP (**LipB**, EC 2.3.1.181).
2. Octanoyl-[lipoyl domain] + 2 S (via 2 SAM, 4Fe-4S) → lipoyl-[lipoyl domain] (**LipA**, EC 2.8.1.8).

The octanoylated protein made by LipB is the **direct macromolecular substrate** of LipA (P 20882995).

Kept separate (neighboring pathways / overview breadth): - Lipoyl-cofactor utilisation: the 2-oxoacid dehydrogenase complexes (PDH aceE/aceF, OGDH sucA/sucB, BCKAD bkdAA/bkdAB/bkdB, acetoin acoC) and the glycine cleavage system (gcvP/gcvH/gcvT). Their lipoyl-domain proteins are *substrates* of the module, not steps within it. - **Exogenous lipoate salvage:** lipoate–protein ligase A (LplA) — explicitly outside the module boundary (and, notably, not annotated in KT2440; no full-length LplA-family protein detected by domain scan). - **Alternative lipoylation architectures:** the *Bacillus*-type LipM (octanoyl-ACP:GcvH transferase) + LipL (GcvH→E2 amidotransferase relay) — outside the boundary. KT2440 does encode a **candidate LipM/LipL-family protein, PP_0423** (see §4/§5), so this architecture cannot be dismissed as absent, but it is out-of-module.

Alternate names / DB definitions: KEGG "Lipoic acid metabolism" (ppu00785); MetaCyc "lipoate biosynthesis and incorporation"; the enzymes are also labelled "lipoyl(octanoyl) transferase" (LipB) and "lipoate synthase / Lip-syn / LS" (LipA). LipB is frequently mis-labelled "lipoate-protein ligase B," which can invite confusion with true salvage ligases (LplA).

---

3. Expected step model

Step	Reaction	Expected player	KT2440 gene	Status
1	Octanoyl-ACP → octanoylated lipoyl-domain protein	LipB-family octanoyltransferase	**lipB / PP_4801 / Q88DM4**	**Covered**
2	Octanoylated → lipoylated protein (S insertion)	LipA-family lipoate synthase	**lipA / PP_4800 / Q88DM5**	**Covered**

No third step exists in this module. The relationship "LipB product is LipA substrate" holds and is satisfied by the adjacent lipA/lipB pair.

4. Candidate genes and evidence

In-module (promote / high confidence)

Gene	Locus / UniProt	Role	EC	Evidence	Family signature	Caveats
lipB	PP_4801 / Q88DM4	Octanoyltransferase (Step 1)	2.3.1.181	UniProt PE3 (homology), score 3/5	PROSITE **LIPB** PS01313; Pfam PF21948; IPR000544	Name "lipoate-protein ligase B" is a misnomer — it is a de novo octanoyltransferase, **not** a salvage ligase.
lipA	PP_4800 / Q88DM5	Lipoyl synthase (Step 2)	2.8.1.8	UniProt PE3 (homology), score 3/5	Radical_SAM PF04055; IPR003698; LIAS_N	Requires 4Fe-4S + SAM; correctly annotated.

Evidence type for both is **homology / rule-based** (no KT2440-specific biochemistry located), but family assignment is unambiguous (exact PROSITE/Pfam diagnostic hits) and genomic adjacency reinforces a functional biosynthetic unit. Transfer of *E. coli* mechanism to KT2440 is **strong** (well-conserved γ-proteobacterial enzymes, correct EC and catalytic-domain signatures).

Authoritative pathway assignment (UniProt/HAMAP). Beyond the KEGG bucket, UniProt PATHWAY comments (HAMAP rules, ECO:0000255) place both enzymes explicitly in *"Protein modification; protein lipoylation via endogenous pathway"* — an exact match to this module's name: - **lipB (HAMAP MF_00013):** catalytic activity `octanoyl-[ACP] + L-lysyl-[protein] = N6-octanoyl-L-lysyl-[protein] + holo-[ACP] + H+` — **Step 1** verbatim. - **lipA (HAMAP MF_00206):** `N6-octanoyl-L-lysyl-[protein] + 2 S (radical-SAM, second [4Fe-4S] cluster) → N6-lipoyl-L-lysyl-[protein]` with a **[4Fe-4S] cofactor** annotation — **Step 2** verbatim.

This is authoritative rule-based confirmation that the two steps are represented by lipA/lipB, independent of KEGG map breadth.

Essentiality (inferred). No KT2440-specific mutant phenotype for lipA/lipB was located; a KT2440 Tn-seq study exists (P 40302248) but targets metal tolerance, not lipoylation. Because no salvage ligase is annotated, de novo lipoylation is the sole route and lipA/lipB are expected to be essential for growth on non-fermentable carbon (PDH/OGDH-dependent) — an inference, not a direct KT2440 result.

Out-of-module bucket members (acceptors and utilisation machinery — do not treat as steps)

- **Lipoyl-domain acceptors (substrates of the module; UniProt 'Lipoyl' keyword):** aceF (PP_0338), sucB (PP_4188), bkdB (PP_4403), acoC (PP_0553), gcvH1 (PP_0989), gcvH2 (PP_5193). These are *products* of lipoylation, not enzymes that install it.
- **Non-lipoylated E1 decarboxylase subunits:** aceE (PP_0339), sucA (PP_4189), bkdAA (PP_4401), bkdAB (PP_4402), gcvP1 (PP_0988), gcvP2 (PP_5192).
- **E3 dihydrolipoyl dehydrogenases (EC 1.8.1.4):** lpd (PP_5366), lpdG (PP_4187, PE1 / 3D-structure — experimentally characterised), lpdV (PP_4404). Use lipoamide but are not lipoyl-domain proteins and not biosynthetic.
- **Glycine-cleavage T-protein aminomethyltransferases (EC 2.1.2.10):** gcvT-I (PP_0986), gcvT (PP_5194).

Paralog notes relevant to the module's substrate load

KT2440 encodes **two glycine-cleavage clusters** (gcvTHP-1 at PP_0986–0989 and gcvTHP-2 at PP_5192–5194) and **three E3s** (lpd/lpdG/lpdV) plus multiple 2-oxoacid E2 acceptors. This expands the population of lysine substrates LipB/LipA must service but does not change module satisfiability.

---

5. Gaps, ambiguities, and likely over-annotations

- **No gap in the module itself.** Both steps are encoded by unambiguous family members.
- **Over-inclusive bucket:** 17/19 kegg:ppu00785 genes are utilisation/acceptor genes, not biosynthesis. This is the main curation risk — a naïve reading would over-count "steps."
- **LipB name ambiguity:** "Lipoate-protein ligase B" wording risks conflation with salvage LplA. The EC (2.3.1.181) and PROSITE LIPB signature confirm octanoyltransferase identity.
- **No annotated salvage ligase (LplA):** neither the name search (only lipB/lipA) nor the domain scan (IPR004143 → lipB, birA, PP_0423) revealed a full-length LplA-family salvage ligase. This is suggestive, not definitive.
- **Uncharacterised LipM/LipL-family protein PP_0423 (Q88QR5):** flagged by domain scan, not by gene name. InterPro IPR050664 "Octanoyltransferase LipM/LipL"; PANTHER PTHR43679:SF2 "octanoyl-[GcvH]:protein N-octanoyltransferase"; PF21948; 234 aa; UniProt PE4 (Predicted), score 1/5, no assigned gene name. It is a candidate GcvH-relay octanoyltransferase (out-of-module) whose real activity is unverified — it could be a functional relay enzyme, a redundant/auxiliary transferase, or an over-called family assignment. Its short single-domain length argues against it being a full LplA. **Genomic context (this review):** PP_0423 is genomically isolated, flanked by tryptophan/pABA-biosynthesis genes (pabA-trpD-trpC, PP_0420–0422) and a crp regulator (PP_0424) — with *no* neighboring gcvH, lipoyl-domain acceptor, or acetoin (aco) gene. There is therefore no operonic support for a dedicated GcvH-relay role, which weakens (but does not exclude) a functional relay interpretation. (A pairwise-alignment attempt to sub-classify it as LipM vs LipL was inconclusive due to reference and local-alignment artifacts; the InterPro/PANTHER family call is the reliable evidence.) This is the single most important item a curator should resolve, because it bears on whether KT2440 has redundant lipoylation routes.

- **acoC (PP_0553)** is UniProt PE4 (Predicted), score 1/5 — lowest-confidence acceptor; not module-critical but flag if it enters an acetoin module.
- **Evidence is homology-level** for lipA/lipB; no KT2440-specific mutant/enzymology found in this pass.

---

6. Module and GO-curation recommendations

Module element	Recommended status
Step 1 (octanoyl transfer, LipB)	**covered** — lipB / PP_4801
Step 2 (sulfur insertion, LipA)	**covered** — lipA / PP_4800
LplA salvage route	**not_expected_in_target_taxon** (outside boundary; no full-length LplA detected)
LipM/LipL (GcvH-relay) alternative	**candidate_uncertain / out_of_module** — PP_0423 is a LipM/LipL-family candidate (PE4); not needed for satisfiability (LipB present) but present and unverified
17 acceptor/E1/E3/GcvT bucket genes	**out_of_module** (belong to 2-oxoacid dehydrogenase / glycine-cleavage modules)

- **Module boundary is correct** for this organism as written (LipB→LipA only). No `module_needs_revision`.
- **GO suggestions:** confirm lipB → GO:0033819 (octanoyltransferase / lipoyl(octanoyl) transferase activity) and lipA → GO:0016992 (lipoate synthase activity); both to GO:0009107 (lipoate biosynthetic process). Acceptors should carry protein-lipoylation *target* terms (e.g. GO:0009249 protein lipoylation) but not the biosynthetic-enzyme activity terms.
- Recommend the local bucket flag lipA/lipB as the only "biosynthetic step" members and demote the other 17 to "cofactor-utilising / substrate" annotations.

---

7. Genes to promote to full `fetch-gene` review

1. **lipB / PP_4801 (Q88DM4)** — the rate-defining, name-ambiguous Step-1 enzyme; confirm octanoyltransferase vs ligase identity and operon context. **High priority.**
2. **lipA / PP_4800 (Q88DM5)** — Step-2 radical-SAM enzyme; confirm 4Fe-4S cofactor and lipA-lipB genomic unit. **High priority.**
3. **PP_0423 (Q88QR5)** — uncharacterised LipM/LipL-family protein flagged by domain scan; resolve whether it is a functional GcvH-relay octanoyltransferase, a redundant transferase, or an over-called family assignment. **High priority for boundary/redundancy decisions** (out-of-module, but its existence affects how the module's exclusions are worded for this taxon).
4. *(Lower priority, only if adjacent modules are opened)* lpdG / PP_4187 (has 3D structure, PE1) as the experimentally-supported E3 anchor; acoC / PP_0553 to resolve its PE4 low-confidence acceptor annotation.

---

8. Key references

- Christensen QH & Cronan JE. "Lipoic acid synthesis: a new family of octanoyltransferases generally annotated as lipoate protein ligases." *Biochemistry* 2010.

  P 20882995

  — defines the LipB→LipA two-step de novo module and the *Bacillus* LipM alternative (basis for boundary and alternative-pathway calls).
- Dhembla et al. "Lipoate protein ligase B primarily recognizes the C-terminal…" 2022.

  P 35764173

  — LipB vs LplA distinction and de novo vs salvage routes.
- Lennox-Hvenekilde et al. "Metabolic engineering of *E. coli* for high-level production of free lipoic acid." *Metab Eng* 2023.

  P 36639019

  — reiterates LipB + LipA as the native de novo enzymes acting on lipoyl domains.
- UniProt entries Q88DM4 (lipB) and Q88DM5 (lipA), *P. putida* KT2440; InterPro IPR000544 / IPR003698; PROSITE PS01313 (LIPB), PS51918 (Radical-SAM); UniProt/HAMAP rules

MF_00013 (lipB) and MF_00206 (lipA) assigning both to "protein lipoylation via endogenous pathway."

- UniProt entry Q88QR5 (PP_0423), *P. putida* KT2440 — "BPL/LPL catalytic domain-containing protein," InterPro IPR050664 (Octanoyltransferase LipM/LipL), PANTHER PTHR43679:SF2 (octanoyl-[GcvH]:protein N-octanoyltransferase); basis for the LipM/LipL relay-candidate call.
- Christensen QH & Cronan JE. "Three transferases...LipM/LipL/GcvH relay" and related B. subtilis lipoylation studies establish the GcvH-relay architecture that PP_0423's family (IPR050664) belongs to. See also

  P 20882995

  above.

Confidence: Module satisfiability call is HIGH (unambiguous family signatures for both steps, genomic adjacency). Enzyme-level evidence is homology-based (UniProt PE3) with strong transfer from E. coli; no KT2440-specific enzymology was located in this pass. Absence of a salvage ligase is annotation-based and should be treated as suggestive.
